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`https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/platformio`

Abstract

This manual explains Karnaugh maps (K-map) by finding the logic functions for the incrementing decoder.

1 Incrementing Decoder

The decoder takes binary inputs from 0 to 9 and outputs the incremented value. The truth table is:

Z	Y	X	W		D	C	B	A
0	0	0	0		0	0	0	1
0	0	0	1		0	0	1	0
0	0	1	0		0	0	1	1
0	0	1	1		0	1	0	0
0	1	0	0		0	1	0	1
0	1	0	1		0	1	1	0
0	1	1	0		1	0	0	1
0	1	1	1		1	0	1	0
1	0	0	0		0	0	0	0

Table 1: Truth table for incrementing decoder

2 Karnaugh Map Minimization

Output A

Boolean expression:

$$A = W'X'Y'Z' + W'XY'Z' + W'X'YZ' + W'XYZ' + W'X'Y'Z$$

Minimized using K-map:

$$A = W'Z' + W'X'Y'$$

Output B

Boolean expression:

$$B = WX'Y'Z' + W'XY'Z' + WX'YZ' + W'XYZ'$$

Minimized:

$$B = WX'Z' + W'XZ'$$

Output C

Boolean expression:

$$C = WXY'Z' + W'X'YZ' + WX'YZ' + W'XYZ'$$

Minimized:

$$C = WXY'Z' + X'YZ' + W'YZ'$$

Output D

Boolean expression:

$$D = WXYZ' + W'X'Y'Z$$

Minimized using K-map directly.

3 Display Decoder

The display decoder maps outputs A, B, C, D to segments a–g. The truth table is:

D	C	B	A	Decimal	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	1	0	1	1	0	0	0	0
0	0	1	0	2	1	1	0	1	1	0	1
0	0	1	1	3	1	1	1	1	0	0	1
0	1	0	0	4	0	1	1	0	0	1	1
0	1	0	1	5	1	0	1	1	0	1	1
0	1	1	0	6	1	0	1	1	1	1	1
0	1	1	1	7	1	1	1	0	0	0	0
1	0	0	0	8	1	1	1	1	1	1	1
1	0	0	1	9	1	1	1	1	0	1	1

Table 2: Truth table for seven segment display decoder

Note